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TITLE

by

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for the Degree of Masters of Science in
Mechanical Engineering
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2026

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ACKNOWLEDGMENTS

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ABSTRACT

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TABLE OF CONTENTS

ACKNOWLEDGMENTS	ii
ABSTRACT	iii
LIST OF TABLES	vi
LIST OF FIGURES	vii
CHAPTER 1 INTRODUCTION	1
CHAPTER 2 ELECTRO-THERMAL HARDWARE-IN-THE-LOOP DIGITAL TWIN UPDATING FOR A GOLDFISH WITH STRONG OPINIONS	3
2.1 Abstract	3
2.2 Introduction	4
CHAPTER 3 ELECTRO-THERMAL HARDWARE-IN-THE-LOOP DIGITAL TWIN UPDATING FOR AN OTTER WITH A LOVE OF TEST BENCHES	5
3.1 Abstract	5
3.2 Introduction	6
CHAPTER 4 ELECTRO-THERMAL HARDWARE-IN-THE-LOOP DIGITAL TWIN UPDATING FOR A HUMMINGBIRD WITH PERFORMANCE ANX- IETY	7
4.1 Abstract	7
4.2 Introduction	8

CHAPTER 5 CONCLUSION	9
BIBLIOGRAPHY	10
APPENDIX A	11
A.1 Chapter 1: Additional Paper	11

LIST OF TABLES

LIST OF FIGURES

Figure 1.1	A Duck doing Duck things.	1
Figure 1.2	A comparison between the required resources, such as time, money, and understanding, to develop each level of battery emulation and their relative benefits.	2
Figure 2.1	The temperature during a) 1C and b) 2C constant current discharges of a K2 26650 LFP cell along with the emulator results for the same cell.	4
Figure 3.1	The temperature during a) 1C and b) 2C constant current discharges of a K2 26650 LFP cell along with the emulator results for the same cell.	6
Figure 4.1	The temperature during a) 1C and b) 2C constant current discharges of a K2 26650 LFP cell along with the emulator results for the same cell.	8

CHAPTER 1

INTRODUCTION

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Figure 1.1 A Duck doing Duck things.

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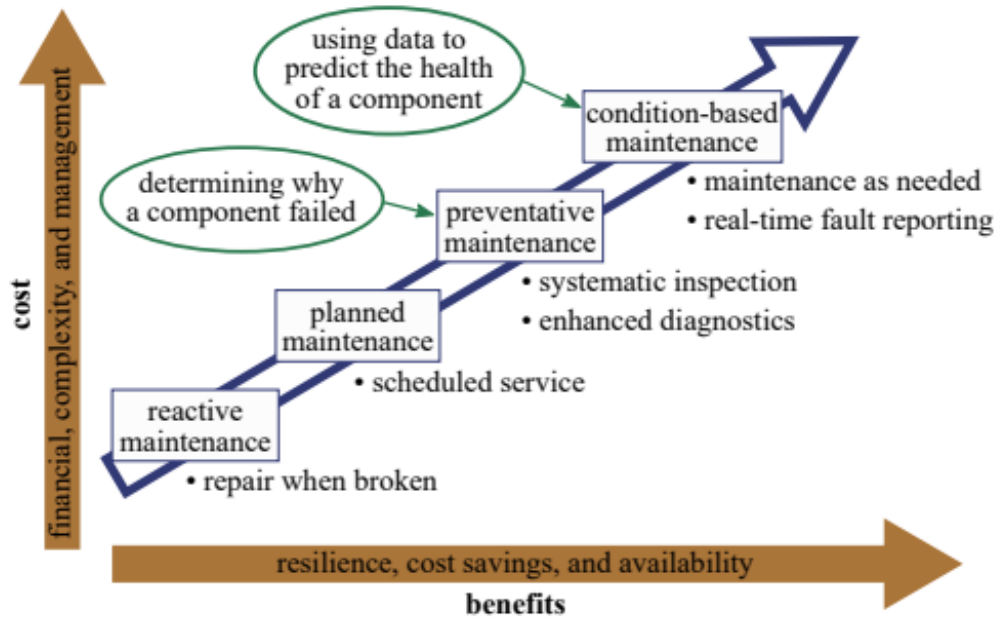


Figure 1.2 A comparison between the required resources, such as time, money, and understanding, to develop each level of battery emulation and their relative benefits.

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CHAPTER 2

ELECTRO-THERMAL HARDWARE-IN-THE-LOOP DIGITAL TWIN UPDATING FOR A GOLDFISH WITH STRONG OPINIONS

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This chapter is wholly based on “Electro-thermal Hardware-in-the-Loop Digital Twin Updating for a Goldfish with Strong Opinions,” accepted for publication in the IEEE ESTS 2025 Conference Proceedings.

2.1 ABSTRACT

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras



Figure 2.1 The temperature during a) 1C and b) 2C constant current discharges of a K2 26650 LFP cell along with the emulator results for the same cell.

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2.2 INTRODUCTION

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

CHAPTER 3

ELECTRO-THERMAL HARDWARE-IN-THE-LOOP DIGITAL TWIN UPDATING FOR AN OTTER WITH A LOVE OF TEST BENCHES

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Figure 3.1 The temperature during a) 1C and b) 2C constant current discharges of a K2 26650 LFP cell along with the emulator results for the same cell.

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3.2 INTRODUCTION

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CHAPTER 4

ELECTRO-THERMAL HARDWARE-IN-THE-LOOP DIGITAL TWIN UPDATING FOR A HUMMINGBIRD WITH PERFORMANCE ANXIETY

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Carolina, Columbia, SC, USA

This chapter is wholly based on “Electro-thermal Hardware-in-the-Loop Digital Twin Updating for a Goldfish with Strong Opinions,” accepted for publication in the IEEE ESTS 2025 Conference Proceedings.

4.1 ABSTRACT

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Figure 4.1 The temperature during a) 1C and b) 2C constant current discharges of a K2 26650 LFP cell along with the emulator results for the same cell.

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4.2 INTRODUCTION

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

CHAPTER 5

CONCLUSION

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APPENDIX A

A.1 CHAPTER 1: ADDITIONAL PAPER

Add more papers as needed.